

Little Man programming, super-detailed: the **Triangle** and **Memory** machines

team wheezards — ICFPC 2026 working notes

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This document explains two of our contest programs at full depth: the tiny **Triangle** machine (rank 1 of 239) and the packed **Memory** machine. Each program is shown twice — as the literal `.man` grid and in our geometry-free *block-graph* notation — and each comes with an informal explanation, a step-by-step walkthrough, and a glossary of exactly the commands it uses. Every number in this document (traces, tick counts, packing arithmetic) was produced by running the real simulator and the real artifacts, not from memory.

1 The machine in ninety seconds

A `.man` program is an ASCII grid. **Rooms** are rectangles drawn with `+ - |` walls. A little man spawns at every `@` inside a room and always starts facing *East*. He carries three registers:

A	main hand	almost every operation reads or writes it
B	off hand	written <i>only</i> by M, W and / (proven empirically)
BP	backpack	a counter; written by b, m; branched on by d

Time advances in ticks. On every tick, in this order: (1) every pipe shifts its values one cell forward; (2) the output room emits a finished value, the input room injects a waiting one; (3) every man *executes the operation he is standing on, then steps one cell in his heading* — unless he is halted or blocked. Two men colliding halt both. *Walking into a wall is an error that fails the whole run* — a correct program never does it (but see the Triangle section for why a program may legally be *about* to do it when the judge stops the clock).

Pipes are directed FIFO conveyors drawn outside rooms with `- |` bodies and arrowheads; each pipe cell holds at most one value, and values travel *one cell per tick* — a pipe of length L is also a delay of L ticks. `s` sends **A** into the *nearest* outgoing pipe and blocks while the pipe’s entry cell is occupied; `r` receives into **A** from the nearest incoming pipe and blocks until a value has arrived at its end. “Nearest” is Manhattan distance from the instruction cell to the point where the pipe crosses the room wall. The special rooms `|I|` and `|O|` are the program’s input and output: the judge feeds values through the **I** room’s pipe and reads answers arriving at **O**.

Numbers. A bare digit sets **A** = 0..9. A backtick literal like `‘123’` loads its number into **A** when the man walks onto the *closing* backtick (so the same cells read differently walked left-to-right and right-to-left). All arithmetic is signed 64-bit.

Halting and scoring. **H** halts one man forever; the run ends when every man has stopped — or earlier, the moment the judge has seen the last expected output. The score of a passing program is

$$\max(\text{width}, \text{height})^2 \times \text{average ticks},$$

lower is better; the bounding box covers the *entire* program including pipes and I/O rooms. Some problems (not these two) are footprint-only.

2 The geometry-free block-graph notation

The `.man` grid mixes two things: the *algorithm* (the sequence of operations the man walks over) and the *geometry* (where corridors turn). Our block-graph notation strips the geometry. A program becomes basic blocks of straight-line operations:

- `(mark L1_8_E)` names a block — auto-generated from the cell (row 1, column 8) and heading (East) where the block starts;
- operations follow in walking order; `(goto L)` is an unconditional jump (in the grid: a corridor);
- `(if NEG ZERO POS)` is the three-way branch of `X` (turn by the sign of `A`);
- `(if-bp TURN STRAIGHT)` is the two-way branch of `d` (turn clockwise if `BP > 0`);
- `(r p21_8) / (s p23_30)` name *which* pipe an `r/s` binds to, by the coordinates of the cell where that pipe crosses the room wall — in the grid this is implicit nearest-pipe resolution, and making it explicit is exactly what a binding audit checks;
- `(wall)` marks a path that would run into a wall — reaching it is an error, so in a correct machine it is dead code.

A note on the bare heading ops (`<^v>`) that still appear inside blocks: every *conditional* turn is lifted into an explicit branch form with absolute targets, and literal values are resolved, so heading carries no control-flow meaning in this notation. The remaining arrows are semantic no-ops kept deliberately, because one op is one tick: a block’s length IS its cost, and every walked cell stays accounted for. Read as a language they are noise; read as a transcript with the geometry erased they are the cost model. A planned normalisation replaces each such run with `(nop n)` — an explicit “*n* ticks pass here” — which keeps the cost model while cleaning the text, and doubles as a compile-side directive for timing-sensitive machines, where exact delays are today built by hand-counting corridor cells.

Both notations below are generated mechanically from the artifacts by our decompiler, and the decompiled form re-validates against the simulator — so what you read is what actually runs.

3 Triangle

The problem. *Output the n -th triangular number: read one integer n , output $1 + 2 + \dots + n$; for $n = 0$ output 0. Six public test cases, one number each. Scoring is footprint $\times$ ticks.*

Two of our machines appear below. `triangle_02.man` (9×9 , 11 ticks, score 891) is the teaching machine: two rooms, two men, three pipes, visible blocking. The actual record holder — rank 1 of about 240 — is its smaller sibling `triangle_03.man` (8×8 , 13 ticks, score $64 \times 13 = \mathbf{832}$), explained in section 3.5: it wins with MORE ticks by exploiting the squared footprint, and it ends by deliberately sacrificing its man.

3.1 The program, in both notations

```
col 012345678
  0 +-----+
  1 |@rM*+sH|
  2 +-----+
  3 +-+>^+-+v
  4 |I| >|O||
  5 +-+ ^+-+v
  6 +-----+
  7 |@1Mr}sH|
  8 +-----+
```

Two rooms (rows 0–2 and rows 6–8), each with one man; the I room feeds the top room through the 2-cell pipe `>^` (row 3, cols 3–4); the top room sends to the bottom room down the 3-cell pipe at column 8 (rows 3–5); the bottom room sends to 0 through the 2-cell pipe `^>` (rows 4–5, col 4). In block-graph form the geometry disappears and each man is one straight line:

```
man 1: (mark L1_1_E) @ r M * + s H
man 2: (mark L7_1_E) @ 1 M r } s H
```

3.2 Glossary — every operation this program uses

- @ the man’s start cell; ordinary space once he walks off it
- r receive: $A \leftarrow$ next value from the nearest incoming pipe; *blocks* until one arrives
- M $B \leftarrow A$ (A unchanged)
- * $A \leftarrow A \times B$
- + $A \leftarrow A + B$
- 1 $A \leftarrow 1$
- } arithmetic shift right: $A \leftarrow A \gg B$ (sign-preserving)
- s send: push A into the nearest outgoing pipe; *blocks* while its entry cell is full
- H halt this man; he stays on the H forever

3.3 The algorithm, informally

The identity being computed is

$$T(n) = \frac{n(n+1)}{2} = \frac{n^2 + n}{2} = (n^2 + n) \gg 1,$$

and the shift is exact because $n^2 + n = n(n+1)$ is a product of consecutive integers, hence always even.

The work is split between two specialists connected by a pipe:

- **man 1** (top room) reads n , computes $n^2 + n$ with three register operations — **M** parks n in **B**, $*$ squares, $+$ adds the parked n — sends the result downstream, halts.
- **man 2** (bottom room) first loads the constant: **1M** puts 1 into **B**. *He does this while man 1 is still computing* — the two run in parallel. Then he blocks on **r** until $n^2 + n$ arrives, halves it with **}** (shift right by $B = 1$), sends the answer to **0**, halts.

Why two rooms at all? A single-room version must fit $\mathbf{rM*+M1W}sH$ plus turns into one box and ends up no smaller, while the two-room split lets each room be a bare one-row corridor and overlaps man 2's constant-loading with man 1's arithmetic. Note also the register discipline: man 2 sets $B=1$ *before* blocking on **r**, because **r** only writes **A** — **B** rides through the wait untouched.

3.4 Step by step: $n = 3$, tick by tick

The table shows the state *after* each tick: where the man stands, the operation now under his feet (executed on the *next* tick), and his registers after this tick's execution. Input: 3. Expected output: 6.

tick	man 1 (top room)					man 2 (bottom room)				
	at	on	A	B		at	on	A	B	
1	(1,2)	r	0	0		(7,2)	1	0	0	
2	(1,3)	M	3	0		(7,3)	M	1	0	
3	(1,4)	*	3	3		(7,4)	r	1	1	
4	(1,5)	+	9	3		(7,4)	r	1	1	
5	(1,6)	s	12	3		(7,4)	r	1	1	
6	(1,7)	H	12	3		(7,4)	r	1	1	
7	(1,7)	H	12	3	halted	(7,4)	r	1	1	
8	(1,7)	H	12	3	halted	(7,5)	}	12	1	
9	(1,7)	H	12	3	halted	(7,6)	s	6	1	
10	(1,7)	H	12	3	halted	(7,7)	H	6	1	
11	(1,7)	H	12	3	halted	(7,7)	H	6	1	halted

Reading it:

- **tick 1–2:** the input value 3 travels the 2-cell input pipe while man 1 walks off **@**; his **r** fires on tick 2 ($A = 3$) with no waiting — the pipe delay was hidden behind his first step. Man 2 meanwhile loads 1 and copies it to **B** (**M** on tick 2).
- **tick 3–5, man 1:** $*$ makes 9, $+$ makes 12, and on tick 5 he stands on **s**.
- **tick 3–7, man 2:** he reached his **r** on tick 3 and *blocks* — position frozen at (7,4) — because nothing has arrived yet. Blocking costs him nothing: he got his constant ready early.
- **tick 6:** man 1 executes **s**: the value 12 enters the 3-cell pipe at column 8. He steps onto **H** and halts on tick 7.
- **tick 6–8:** 12 rides the pipe one cell per tick and reaches the pipe's end; man 2's **r** fires on tick 8 ($A = 12$, B still 1).
- **tick 9–10:** **}** halves: $A = 6$; **s** sends 6 into the output pipe.
- **tick 11:** 6 arrives at **0**. The judge has now seen every expected output and *stops the clock*: 11 ticks. (Man 2 halts on his **H** the same tick; had he been about to walk into a wall instead, it would not have mattered — ticks after the last output never run.)

Score: bounding box $9 \times 9 \Rightarrow 81$; ticks 11; $81 \times 11 = 891$. The lesson that generalises: *overlap work across men, park constants in B before blocking, and remember the run is only as long as the judge needs it to be.*

3.5 The record holder: one room, no halt, a sacrificed man

`triangle_03.man` beats the machine above 891 $\rightarrow$ 832 while taking *more* ticks (13 vs 11) — because the score squares the bounding box: $64 \times 13 = 832 < 81 \times 11 = 891$. Losing one column and one row was worth two extra ticks. Its other trick is more radical: it contains **no H at all**.

```
col 01234567
  0 +-----+
  1 |@rM*+v|
  2 |      M|
  3 | s}W1<|
  4 +-----+
  5 ^<+--+v
  6   |I|O|<
  7   +--++
```

```
(mark L1_1_E) @ r M * + v M < 1 W } s (goto __wall)
(mark __wall) (wall)
```

The block-graph ends in `(goto __wall) (wall)`: after `s`, the man walks over a blank cell and *into the west wall*. In the raw simulator that run's status is literally **error** — and the case still passes, because the answer reaches 0 on the very tick of the fatal step, and the server accepts a run whose final tick both delivers the last output and steps into a wall (a behaviour we probed and encoded in `server_compat` as final-wall-step semantics). The H cell, and the room width to reach it, are simply deleted.

Trace for $n = 3$ (single man; same column conventions as above):

tick	at	on	A	B
1	(1,2)	r	0	0
2	(1,3)	M	3	0
3	(1,4)	*	3	3
4	(1,5)	+	9	3
5	(1,6)	v	12	3
6	(2,6)	M	12	3
7	(3,6)	<	12	12
8	(3,5)	1	12	12
9	(3,4)	W	1	12
10	(3,3)	}	12	1
11	(3,2)	s	6	1
12	(3,1)		6	1
13	(3,1)		6	1

Ticks 1–5 are the same $n^2 + n$ prologue as before, ending with `v` into row 2, whose lone `M` parks $n^2 + n$ in B in passing; row 3 walked westward reads `<1W}s`: load 1, swap, halve, send. Ticks 12–13: the man crosses the blank cell and dies on the wall exactly as 6 arrives at 0. Three lessons: *the footprint is squared, so trade ticks for cells; a man you no longer need costs nothing to kill; and the scoring clock, not the man's fate, decides the run* — but note the sibling `triangle_04` fails 0/6 from getting exactly this timing wrong, so the sacrifice must be *proven*, not eyeballed.

4 Memory

The problem. *Simulate a 100-cell memory. The input is a stream of operations: a READ is the pair 0 **addr**; a WRITE is the triple 1 **addr value**. Every cell starts at 0. For each READ, output the current value at **addr**; a WRITE produces no output. Example from the problem statement: the stream 0 5 1 5 10 1 6 9 0 6 0 5 outputs 0 9 10. Values fit in $[-10^6, 10^6]$; scoring is footprint $\times$ ticks.*

Our machine here is `submissions/memory/memory_04.man` (37×37 , 24/24 on the server), generated by `src/littleman/memory_packed.py`. It is presented because every mechanism below is documented and asserted by that generator’s tests. (The team’s best live Memory score is a sibling, hand-tuned lineage — `alexey-memory_06`, 34×32 — built on the same ideas.)

4.1 The two ideas

Idea 1: memory as a circulating ring. A littleman machine has no random access — but a pipe is a FIFO, and two pipes plus two rooms make a *ring* on which the whole memory circulates. To touch address k positions ahead of the ring’s head you relay k items from the incoming ring pipe to the outgoing one; each relay costs ~ 9 ticks. The naive version circulates all 100 cells; profiling it showed ring relays were 93% of all ticks. That measurement is what motivated idea 2.

Idea 2: pack three cells per word. A value in $[-10^6, 10^6]$ fits a signed 21-bit field (from $-1\,048\,576$ up to $1\,048\,575$). Three fields live in bits $[0, 21)$, $[21, 42)$, $[42, 63)$ of one signed-64 word:

$$word = (f_0 \wedge M) \vee ((f_1 \wedge M) \ll 21) \vee ((f_2 \wedge M) \ll 42),$$

with $M = 2^{21} - 1$. The ring shrinks from 100 items to $\lceil 100/3 \rceil = 34$, and the relay count on the profiled case drops $3.06\times$. Bit 63 is always zero, so no arithmetic ever wraps, and a freshly zeroed memory is simply the word 0 — the ring seeds with 34 plain zeros instead of long literals.

Address a lives in word $\lfloor a/3 \rfloor$, field $a \bmod 3$ — both produced by *one* / operation (which leaves the quotient in **A** and the remainder in **B**). With $shift = 21 \cdot (a \bmod 3)$:

$$\begin{array}{ll} \text{read:} & v = (word \ll (43 - shift)) \gg 43 \\ \text{write:} & word' = (word \wedge \neg(M \ll shift)) \vee ((v \wedge M) \ll shift) \end{array}$$

The read needs no mask and no bias: the left shift parks the field’s sign bit in bit 63, and the *arithmetic* right shift sign-extends it on the way down. Worked numbers, computed with the real formulas:

write 10 at addr 5	word 1, field 2, shift 42	word becomes 43 980 465 111 040 = 10 $\ll$ 42
read addr 5	(<i>word</i> $\ll$ 1) $\gg$ 43	= 10
write -7 at addr 3	word 1, field 0, shift 0	field bits $(-7) \wedge M = 2\,097\,145$
read addr 3	(<i>word</i> $\ll$ 43) $\gg$ 43	= -7 (sign restored by the shift)
read addr 5 again		still 10 — the write of -7 touched only field 0

4.2 Architecture: five rooms in a pipeline

```
I -> P1 -> HEAD -> P2 -> STATION -> 0
                                STATION <-> RELAY      (the 34-word ring)
```

room	grid position	man	job
P1	rows 0–4	(1, 8)	parse one op; split addr into word+shift
HEAD	rows 7–10	(8, 8)	turn word index into ring distance k ; advance head
P2	rows 13–19	(14, 8)	precompute the constants the station needs
STATION	rows 22–36	(23, 7)	owns the ring; does the actual read/write
RELAY	rows 30–33 (right)	(31, 32)	closes the ring (6-tick lap)

Each stage passes a small integer stream to the next; all five men run concurrently, so the pipeline overlaps parsing of the next operation with ring work for the current one.

4.3 Glossary — the additional operations Memory uses

0–9	$A \leftarrow \text{digit}$	'123'	$A \leftarrow \text{literal at closing backtick}$
W	swap $A \leftrightarrow B$	N	$A \leftarrow -A$
-	$A \leftarrow A - B$	/	$A \leftarrow \lfloor A/B \rfloor$, remainder $\rightarrow B$
%	$A \leftarrow A \bmod B$	&	$A \leftarrow A \wedge B$ (bitwise AND)
	$A \leftarrow A \vee B$ (OR)	~	$A \leftarrow A \oplus B$ (XOR)
{	$A \leftarrow A \ll B$	}	$A \leftarrow A \gg B$ (arithmetic)
b	$BP \leftarrow A$	m	$BP \leftarrow BP - 1$
d	turn clockwise if $BP > 0$, else straight	X	turn by sign of A (CW/straight/CCW)
^v<>	set heading N/S/W/E	r s M * + H	as in Triangle

Two facts do most of the planning work here: B is written *only* by M , W , $/$ — so a value parked in B survives any amount of r/s relaying — and BP is a countdown register whose d -branch turns straight-line corridors into loops.

4.4 Stage by stage, with block-graphs

P1 — parse and pre-split.

```
(mark L1_8_E) @ r s b r M 3 W / s s W M '21' * s M (if-bp L2_28_S L1_29_E)
(mark L1_29_E) 0 s W s v v (goto L3_33_S)
(mark L2_28_S) > r s W s v (goto L3_33_S)
(mark L3_33_S) < ^ ^ > (goto L1_8_E)
```

Reads the tag (0=READ, 1=WRITE), forwards it, and *stashes it in the backpack* (b) so one ($\text{if-bp} \dots$) at the end branches the whole room. Reads the address, and $M\ 3\ W\ /$ computes $a/3$ in one stroke: quotient w in A , remainder (the field index) in B . It emits w twice (HEAD needs it once for the distance, once to advance the head), then $\text{shift} = 21 \cdot \text{field}$ via $W\ M\ '21'\ *$. The WRITE arm forwards the value x ; the READ arm substitutes 0 — so downstream always sees the same shape: $[tag, w, w, \text{shift}, x, \text{shift}]$.

HEAD — ring distance, and a register trick.

```
(mark L8_8_E) @ M r s r - M '34' W % s r M 1 + M '34' W % v < M r s r s r s W ^ > (goto L8_8_E)
```

The ring rotates as it is used, so the station never needs an absolute position — only the *distance* $k = (w - hw) \bmod 34$ from the current head word hw . HEAD computes that with $-M\ '34'\ W\ \%$, then the new head $hw' = (w + 1) \bmod 34$. The trick: hw lives in A *across laps* — no storage pipe at all. When the tail $[\text{shift}, x, \text{shift}]$ must be relayed (three r/s pairs, which clobber A), the man first parks hw' in B with M , relays, and takes it back with W — legal precisely because relaying never writes B .

P2 — precompute the station's constants.

```
(mark L14_8_E) @ r s b r s (if-bp L15_14_S L14_15_E)
(mark L14_15_E) r M '43' - s r r v < ^ ^ (goto L16_7_N)
(mark L15_14_S) v > r M '2097151' { M 1 v < N ~ s '2097151' M r & M r W { s ^ (goto L16_7_N)
(mark L16_7_N) ^ ^ > (goto L14_8_E)
```

The station should do as little arithmetic as possible while it holds the ring, so P2 does it in advance. For a READ it sends $[0, k, 43 - \text{shift}]$ (and discards the unused x). For a WRITE it builds both write-formula constants: $'2097151'\ \{$ makes $M \ll \text{shift}$, and the pair $N\ \sim$ turns it into $\neg(M \ll \text{shift})$ — load -1 (all ones), XOR. Then $(x \wedge M) \ll \text{shift}$ via $\&\ \{$. It sends $[1, k, \neg(M \ll s), (x \wedge M) \ll s]$.

STATION — seed, relay, read or splice.

```
(mark L23_7_E) @ '33' b 0 > (goto L23_16_E)
(mark L23_16_E) (s p23_30) v (if-bp L24_16_W L25_17_S)
(mark L24_16_W) m ^ > (goto L23_16_E)
(mark L25_10_N) 3 (goto __wall)
(mark L25_17_S) < v > (goto L26_9_E)
(mark L26_9_E) (r p21_8) (if L25_10_N L26_11_E L27_10_S)
(mark L26_11_E) (r p21_8) b (r p21_8) M v > (goto L27_21_E)
(mark L27_10_S) v v > (r p21_8) (if L29_12_N L30_13_E L31_12_S)
(mark L27_21_E) (r p24_30) (s p23_30) v (if-bp L28_22_W L29_23_S)
(mark L28_7_N) ^ ^ > > (goto L26_9_E)
(mark L28_22_W) m ^ > (goto L27_21_E)
(mark L29_12_N) b b (goto __wall)
(mark L29_23_S) < { M '43' W } (s p35_5) ^ (goto L28_7_N)
(mark L30_13_E) (r p21_8) M v < < (goto L33_22_W)
(mark L31_12_S) > b m (r p21_8) M > (goto L31_21_E)
(mark L31_21_E) (r p24_30) (s p23_30) v (if-bp L32_22_W L33_23_S)
(mark L32_22_W) m ^ > (goto L31_21_E)
(mark L33_22_S) (r p24_30) & M (r p21_8) | v > (s p23_30) v < ^ ^ ^ ^ ^ ^ ^ ^ (goto L28_7_N)
(mark L33_23_S) < (goto L33_22_W)
(mark __wall) (wall)
```

Reading this graph top to bottom:

- **Seed** (L23_7_E): '33' b 0 then a send-decrement loop — 34 zeros into the ring, i.e. the whole zeroed memory in three cells of program text.
- **Dispatch** (L26_9_E): read the tag, (if NEG ZERO POS) — 0 goes to the READ arm, 1 to the WRITE arm. (The NEG arm runs into (wall): dead code, since tags are never negative.)
- **READ arm**: b loads k into the backpack; M parks the decode constant $43 - \text{shift}$ in B; then the relay loop (r ring-in)(s ring-out)(if-bp m,loop | exit) runs $k+1$ times. The subtlety: the *target word itself* is also received *and re-sent* — the ring stays intact — and because r leaves the received word in A, the man exits the loop still holding it. Decode is then {M '43' W}: shift left by $43 - \text{shift}$ (parked in B through the whole relay loop), shift right by 43, and the sign-extended field goes to 0.
- **WRITE arm**: b m sets the countdown to $k - 1$, the mask complement is parked in B, and the loop relays k words unchanged. The $(k+1)$ -th word — the target — is received but *not* re-sent: & clears the field, | splices in the prebuilt $(x \wedge M) \ll \text{shift}$, and only the patched word is sent back to the ring. No output is produced, matching the spec.
- Both arms return to the dispatch block; the station is ready for the next operation while upstream rooms are already parsing it.

RELAY — closing the ring.

```
(mark L31_32_E) @ > (goto L31_34_E)
(mark L31_34_E) r v < s ^ > (goto L31_34_E)
```

A pipe may not start and end at the same room, so the ring needs a second room: a minimal $r v < s ^ >$ lap, 6 ticks per word. Its man runs concurrently with the station's, so the ring keeps flowing while the station computes.

4.5 Step by step: the specification's own example

Feeding the example stream 0 5 1 5 10 1 6 9 0 6 0 5 to the real artifact produces exactly the specified output 0 9 10, at ticks 264, 1011 and 1305. Walking the first two operations through the pipeline, with the packing state:

1. **0 5** (READ 5): P1 splits $5 \rightarrow w=1$, field 2, $shift=42$; HEAD (head $hw=0$) makes $k = (1 - 0) \bmod 34 = 1$ and advances the head to 2; P2 sends $[0, 1, 43-42=1]$. The station relays $k+1 = 2$ words — word 0 (now the ring's tail) and word 1, keeping word 1 in **A**: it is 0, so $(0 \ll 1) \gg 43 = 0$ is sent to 0. Output **0** at tick 264.
2. **1 5 10** (WRITE 10 at 5): same split ($w=1$, shift 42), but the head is now 2, so $k = (1 - 2) \bmod 34 = 33$ — the ring must rotate almost a full turn, which is why this operation is the expensive one (~ 747 ticks: 33 relays at ~ 9 ticks each, plus the pipeline). P2 sends $\neg(M \ll 42)$ and $10 \ll 42 = 43\,980\,465\,111\,040$; the station relays 33 words, patches word 1 to exactly that value, and sends it back. No output.
3. The remaining three operations land the outputs **9** (read of cell 6, field 0 of word 2) and **10** (read of cell 5 again) — the packed word round-trips both fields independently, which is the whole point of the mask-and-splice write.

The cost model this exposes — *an operation costs its ring distance* — is also the machine's weakness: an adversarial address pattern (alternating between two cells 17 words apart) forces half-turns forever. Packing bought a flat $3\times$ on exactly that cost, which is why it was worth the extra arithmetic in P2.

4.6 The full program

```
col 0123456789012345678901234567890123456
0      +-----+
1  +-+  |>@rsbrM3W/ssWM'21'*sMd0sWsv|
2  |I|>->|^                                >rsWsv|
3  +-+  |^                                <|
4      +-----+
5          v
6          v
7      +-----+
8  |>@Mrsr-M'34'W%srM1+M'34'W%v|
9  |^                                WsrsrM<|
10     +-----+
11         v
12         v
13     +-----+
14 |>@rsbrsdrM '43'-srr    v|
15 |^                v    |
16 |^                >rM'2097151'{M1v |
17 |^s{WrM&rM'1517902' s^N< |
18 |^                                <|
19     +-----+
20         v
21         v
22     +-----+
23 |@'33'b0 >sv                |>----v
24 |                ^md        |<    |
25 | v                <        ||    |
26 |>>rXrbrM          v        ||v---<
27 |^ v                >rsv     |||
28 |^ v                ^ md     ||>----v
29 |^s                }W'34'M{<  || v--<
30 |^ >rXrM            v        ||+----+
31 |^ >bmrM            >rsv     |||@>rv|
32 |^                ^ md     ||| ^s<|
33 |^ v|r            M&r< <    ||+----+
34 +-+  |^ >                sv ||    v
35 |0|<-<|^                < |^---<
36 +-+  +-----+
```

Score: bounding box $37 \times 37 \Rightarrow 1369$; server average $\sim 20\,270$ ticks over 24/24 cases; score 27 753 851.

5 Where everything lives, and how this document was made

<code>submissions/triangle/triangle_02.man</code>	the live Triangle artifact (rank 1/239)
<code>submissions/memory/memory_04.man</code>	the packed Memory artifact explained here
<code>src/littleman/memory_packed.py</code>	Memory generator; the packing formulas with tests
<code>src/littleman/sim.py</code>	the reference simulator (the semantics authority)
<code>src/littleman/fastsim.py</code>	the fast executor, proven bit-identical to <code>sim.py</code>
<code>src/littleman/decompile.py</code>	<code>.man</code> → block-graph; round-trips the whole corpus
<code>src/littleman/blockgraph.py</code>	the block-graph parser/interpreter (27 tests)
<code>docs/language-reference.md</code>	the full operation table, verbatim from the contest
<code>docs/architecture/claude_effects.json</code>	the <i>empirical</i> op-effect table (who writes B, etc.)

Every grid, block-graph, trace row and number in this document was generated from those artifacts by the real toolchain at build time — the traces by instrumenting `sim.py` tick by tick, the block-graphs by `decompile.py`, the packing arithmetic by executing the formulas — so if it is written here, it ran.

To re-verify the headline facts yourself:

```
uv run python -m littleman submissions/triangle/triangle_02.man triangle
uv run python -m littleman submissions/memory/memory_04.man memory
uv run pytest tests/test_memory_packed.py -q
```